

14.2 UNSCENTED TRANSFORMATIONS

The problem with nonlinear systems is that it is difficult to transform a probability density function through a general nonlinear function. In the previous section, we were able to obtain exact nonlinear transformations of the mean and covariance, but only for a simple two-dimensional transformation. The extended Kalman filter works on the principle that a linearized transformation of means and covariances is approximately equal to the true nonlinear transformation, but we saw in the previous section that the approximation could be unsatisfactory.

An unscented transformation is based on two fundamental principles. First, it is easy to perform a nonlinear transformation on a single point (rather than an entire pdf). Second, it is not too hard to find a set of individual points in state space whose sample pdf approximates the true pdf of a state vector.

Taking these two ideas together, suppose that we know the mean $\bar{x}$ and covariance P of a vector x . We then find a set of deterministic vectors called sigma points whose ensemble mean and covariance are equal to $\bar{x}$ and P . We next apply our known nonlinear function $y = h(x)$ to each deterministic vector to obtain transformed vectors. The ensemble mean and covariance of the transformed vectors will give a good estimate of the true mean and covariance of y . This is the key to the unscented transformation.

As an example, suppose that x is an $n \times 1$ vector that is transformed by a nonlinear function $y = h(x)$. Choose $2n$ sigma points $x^{(i)}$ as follows:

$$\begin{aligned} x^{(i)} &= \bar{x} + \tilde{x}^{(i)} & i = 1, \dots, 2n \\ \tilde{x}^{(i)} &= \left(\sqrt{nP} \right)_i^T & i = 1, \dots, n \\ \tilde{x}^{(n+i)} &= - \left(\sqrt{nP} \right)_i^T & i = 1, \dots, n \end{aligned} \quad (14.29)$$

where $\sqrt{nP}$ is the matrix square root of nP such that $(\sqrt{nP})^T \sqrt{nP} = nP$, and $(\sqrt{nP})_i$ is the i th row of $\sqrt{nP}$.¹ In the next couple of subsections, we will see how the ensemble mean of the above sigma points can be used to approximate the mean and covariance of a nonlinearly transformed vector.

14.2.1 Mean approximation

Suppose that we have a vector x with a known mean $\bar{x}$ and covariance P , a nonlinear function $y = h(x)$, and we want to approximate the mean of y . We propose transforming each individual sigma point of Equation (14.29) using the nonlinear function $h(\cdot)$, and then taking the weighted sum of the transformed sigma points to approximate the mean of y . The transformed sigma points are computed as follows:

$$y^{(i)} = h \left(x^{(i)} \right) \quad i = 1, \dots, 2n \quad (14.30)$$

¹MATLAB's Cholesky factorization routine CHOL can be used to find a matrix square root. See Section 6.3.1, but note the slight difference between the matrix square root definition used in that section and here.

The true mean of y is denoted as $\bar{y}$. The approximated mean of y is denoted as $\bar{y}_u$ and is computed as follows:

$$\bar{y}_u = \sum_{i=1}^{2n} W^{(i)} y^{(i)} \quad (14.31)$$

The weighting coefficients $W^{(i)}$ are defined as follows:

$$W^{(i)} = \frac{1}{2n} \quad i = 1, \dots, 2n \quad (14.32)$$

Equation (14.31) can therefore be written as

$$\bar{y}_u = \frac{1}{2n} \sum_{i=1}^{2n} y^{(i)} \quad (14.33)$$

Now let's compute the value of $\bar{y}_u$ to see how well it matches the true mean of y . To do this we first use Equation (1.89) to expand each $y^{(i)}$ in Equation (14.33) in a Taylor series around $\bar{x}$. This results in

$$\begin{aligned} \bar{y}_u &= \frac{1}{2n} \sum_{i=1}^{2n} \left(h(\bar{x}) + D_{\bar{x}^{(i)}} h + \frac{1}{2!} D_{\bar{x}^{(i)}}^2 h + \dots \right) \\ &= h(\bar{x}) + \frac{1}{2n} \sum_{i=1}^{2n} \left(D_{\bar{x}^{(i)}} h + \frac{1}{2!} D_{\bar{x}^{(i)}}^2 h + \dots \right) \end{aligned} \quad (14.34)$$

Now notice that for any integer $k \geq 0$ we have

$$\begin{aligned} \sum_{j=1}^{2n} D_{\bar{x}^{(j)}}^{2k+1} h &= \sum_{j=1}^{2n} \left[\left(\sum_{i=1}^n \tilde{x}_i^{(j)} \frac{\partial}{\partial x_i} \right)^{2k+1} h(x) \Big|_{x=\bar{x}} \right] \\ &= \sum_{j=1}^{2n} \left[\sum_{i=1}^n \left(\tilde{x}_i^{(j)} \right)^{2k+1} \frac{\partial^{2k+1} h(x)}{\partial x_i^{2k+1}} \Big|_{x=\bar{x}} \right] \\ &= \sum_{i=1}^n \left[\sum_{j=1}^{2n} \left(\tilde{x}_i^{(j)} \right)^{2k+1} \frac{\partial^{2k+1} h(x)}{\partial x_i^{2k+1}} \Big|_{x=\bar{x}} \right] \\ &= 0 \end{aligned} \quad (14.35)$$

because from Equation (14.29) $\tilde{x}^{(j)} = -\tilde{x}^{(n+j)}$ ($j = 1, \dots, n$). Therefore, all of the odd terms in Equation (14.34) evaluate to zero and we have

$$\begin{aligned} \bar{y}_u &= h(\bar{x}) + \frac{1}{2n} \sum_{i=1}^{2n} \left(\frac{1}{2!} D_{\bar{x}^{(i)}}^2 h + \frac{1}{4!} D_{\bar{x}^{(i)}}^4 h + \dots \right) \\ &= h(\bar{x}) + \frac{1}{2n} \sum_{i=1}^{2n} \frac{1}{2!} D_{\bar{x}^{(i)}}^2 h + \\ &\quad \frac{1}{2n} \sum_{i=1}^{2n} \left(\frac{1}{4!} D_{\bar{x}^{(i)}}^4 h + \frac{1}{6!} D_{\bar{x}^{(i)}}^6 h + \dots \right) \end{aligned} \quad (14.36)$$

Now look at the second term on the right side of the above equation:

$$\begin{aligned}
 \frac{1}{2n} \sum_{i=1}^{2n} \frac{1}{2!} D_{\tilde{x}^{(i)}}^2 h &= \frac{1}{2n} \sum_{k=1}^{2n} \frac{1}{2!} \left(\sum_{i=1}^n \tilde{x}_i^{(k)} \frac{\partial}{\partial x_i} \right)^2 h(x) \Big|_{x=\tilde{x}} \\
 &= \frac{1}{4n} \sum_{k=1}^{2n} \sum_{i,j=1}^n \tilde{x}_i^{(k)} \tilde{x}_j^{(k)} \frac{\partial^2}{\partial x_i \partial x_j} h(x) \Big|_{x=\tilde{x}} \\
 &= \frac{1}{4n} \sum_{i,j=1}^n \sum_{k=1}^{2n} \tilde{x}_i^{(k)} \tilde{x}_j^{(k)} \frac{\partial^2}{\partial x_i \partial x_j} h(x) \Big|_{x=\tilde{x}} \\
 &= \frac{1}{2n} \sum_{i,j=1}^n \sum_{k=1}^n \tilde{x}_i^{(k)} \tilde{x}_j^{(k)} \frac{\partial^2}{\partial x_i \partial x_j} h(x) \Big|_{x=\tilde{x}} \quad (14.37)
 \end{aligned}$$

where we have again used the fact from Equation (14.29) that $\tilde{x}^{(k)} = -\tilde{x}^{(k+n)}$ ($k = 1, \dots, n$). Substitute for $\tilde{x}_i^{(k)}$ and $\tilde{x}_j^{(k)}$ from Equation (14.29) in the above equation to obtain

$$\begin{aligned}
 \frac{1}{2n} \sum_{i,j=1}^n \sum_{k=1}^n \tilde{x}_i^{(k)} \tilde{x}_j^{(k)} \frac{\partial^2 h(x)}{\partial x_i \partial x_j} \Big|_{x=\tilde{x}} &= \frac{1}{2n} \sum_{i,j=1}^n \sum_{k=1}^n (\sqrt{nP})_{ki} (\sqrt{nP})_{kj} \frac{\partial^2 h(x)}{\partial x_i \partial x_j} \Big|_{x=\tilde{x}} \\
 &= \frac{1}{2n} \sum_{i,j=1}^n n P_{ij} \frac{\partial^2 h(x)}{\partial x_i \partial x_j} \Big|_{x=\tilde{x}} \\
 &= \frac{1}{2} \sum_{i,j=1}^n P_{ij} \frac{\partial^2 h(x)}{\partial x_i \partial x_j} \Big|_{x=\tilde{x}} \quad (14.38)
 \end{aligned}$$

Equation (14.36) can therefore be written as

$$\begin{aligned}
 \bar{y}_u &= h(\tilde{x}) + \frac{1}{2} \sum_{i,j=1}^n P_{ij} \frac{\partial^2 h}{\partial x_i \partial x_j} \Big|_{x=\tilde{x}} + \\
 &\quad \frac{1}{2n} \sum_{i=1}^{2n} \left(\frac{1}{4!} D_{\tilde{x}^{(i)}}^4 h + \frac{1}{6!} D_{\tilde{x}^{(i)}}^6 h + \dots \right) \quad (14.39)
 \end{aligned}$$

Now recall that the true mean of y is given by Equation (14.15) as

$$\bar{y} = h(\tilde{x}) + \frac{1}{2!} E [D_{\tilde{x}}^2 h] + \frac{1}{4!} E [D_{\tilde{x}}^4 h] + \dots \quad (14.40)$$

Look at the second term on the right side of the above equation. It can be written as follows:

$$\begin{aligned}
 \frac{1}{2!} E [D_{\tilde{x}}^2 h] &= \frac{1}{2!} E \left[\left(\sum_{i=1}^n \tilde{x}_i \frac{\partial}{\partial x_i} \right)^2 h(x) \Big|_{x=\tilde{x}} \right] \\
 &= \frac{1}{2!} E \left[\sum_{i,j=1}^n \tilde{x}_i \tilde{x}_j \frac{\partial^2 h}{\partial x_i \partial x_j} \Big|_{x=\tilde{x}} \right]
 \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2!} \sum_{i,j=1}^n E(\tilde{x}_i \tilde{x}_j) \frac{\partial^2 h}{\partial x_i \partial x_j} \Big|_{x=\bar{x}} \\
&= \frac{1}{2!} \sum_{i,j=1}^n P_{ij} \frac{\partial^2 h}{\partial x_i \partial x_j} \Big|_{x=\bar{x}}
\end{aligned} \tag{14.41}$$

We therefore see that $\bar{y}$ can be written from Equation (14.40) as

$$\begin{aligned}
\bar{y} &= h(\bar{x}) + \frac{1}{2} \sum_{i,j=1}^n P_{ij} \frac{\partial^2 h}{\partial x_i \partial x_j} \Big|_{x=\bar{x}} + \\
&\quad \frac{1}{4!} E[D_{\bar{x}}^4 h] + \frac{1}{6!} E[D_{\bar{x}}^6 h] + \dots
\end{aligned} \tag{14.42}$$

Comparing this with Equation (14.39) we see that $\bar{y}_u$ (the approximated mean of y) matches the true mean of y correctly up to the third order, whereas linearization only matches the true mean of y up to the first order (see Section 14.1.1). If we compute $\bar{y}_u$ using Equations (14.29), (14.30), and (14.33), then the value of $\bar{y}_u$ will match the true mean of y up to the third order. The biggest difficulty with this algorithm is the matrix square root that is required in Equation (14.29). But the unscented transformation has the computational advantage that the linearization matrix H does not need to be computed. Of course, the greatest advantage of the unscented transformation (relative to linearization) is the increased accuracy of the mean transformation.

14.2.2 Covariance approximation

Now suppose that we want to approximate the covariance of the nonlinearly transformed vector x . That is, we have an n -element vector x with known mean $\bar{x}$ and covariance P , and we have a known nonlinear function $y = h(x)$. We want to estimate the covariance of y . We will denote the estimate as P_u , and we propose using the following equation:

$$\begin{aligned}
P_u &= \sum_{i=1}^{2n} W^{(i)} (y^{(i)} - y_u)(y^{(i)} - y_u)^T \\
&= \frac{1}{2n} \sum_{i=1}^{2n} (y^{(i)} - y_u)(y^{(i)} - y_u)^T
\end{aligned} \tag{14.43}$$

where the $y^{(i)}$ vectors are the transformed sigma points that were computed in Equation (14.30), and the $W^{(i)}$ weighting coefficients are the same as those given in Equation (14.32). Expanding this approximation using Equations (1.89) and (14.36) gives the following:

$$P_u = \frac{1}{2n} \sum_{i=1}^{2n} [h(x^{(i)}) - y_u] [h(x^{(i)}) - y_u]^T \tag{14.44}$$

$$\begin{aligned}
&= \frac{1}{2n} \sum_{i=1}^{2n} \left[h(\bar{x}) + D_{\bar{x}^{(i)}} h + \frac{1}{2} D_{\bar{x}^{(i)}}^2 h + \frac{1}{3!} D_{\bar{x}^{(i)}}^3 h + \dots \right. \\
&\quad \left. - h(\bar{x}) - \frac{1}{2n} \sum_{j=1}^{2n} \left(\frac{1}{2} D_{\bar{x}^{(j)}}^2 h + \frac{1}{4!} D_{\bar{x}^{(j)}}^4 h + \dots \right) \right] \left[\dots \right]^T \quad (14.45)
\end{aligned}$$

Multiplying this equation out gives

$$\begin{aligned}
P_u &= \frac{1}{2n} \sum_{i=1}^{2n} \left\{ (D_{\bar{x}^{(i)}} h) (\dots)^T + \underbrace{\left[\left(\frac{1}{2} D_{\bar{x}^{(i)}} h \right) (D_{\bar{x}^{(i)}}^2 h)^T \right]}_0 + \underbrace{\left[\dots \right]^T}_0 + \right. \\
&\quad \frac{1}{4!} (D_{\bar{x}^{(i)}}^2 h) (\dots)^T - \underbrace{\left[D_{\bar{x}^{(i)}} h \left(\frac{1}{2n} \sum_j \frac{1}{2} D_{\bar{x}^{(j)}}^2 h \right)^T \right]}_0 - \underbrace{\left[\dots \right]^T}_0 + \\
&\quad \frac{1}{4n^2} \left(\sum_j D_{\bar{x}^{(j)}}^2 h \right) (\dots)^T - \left[\frac{1}{4n} D_{\bar{x}^{(i)}}^2 h \left(\sum_j D_{\bar{x}^{(j)}}^2 h \right)^T \right] - \left[\dots \right]^T + \\
&\quad \left. \left[D_{\bar{x}^{(i)}} h \left(\frac{1}{3!} D_{\bar{x}^{(j)}}^3 h \right)^T \right] + \left[\dots \right]^T + \dots \right\} \quad (14.46)
\end{aligned}$$

Some of the terms in the above equation are zero as noted above because $\tilde{x}^{(i)} = -\tilde{x}^{(i+n)}$ for $i = 1, \dots, n$. So the covariance approximation can be written as

$$P_u = \frac{1}{2n} \sum_{i=1}^{2n} (D_{\bar{x}^{(i)}} h) (\dots)^T + \text{HOT} \quad (14.47)$$

where HOT means higher-order terms (i.e., terms to the fourth power and higher). Expanding this equation for P_u while neglecting the higher order terms gives

$$P_u = \frac{1}{2n} \sum_{i=1}^{2n} \sum_{j,k=1}^n \left(\tilde{x}_j^{(i)} \frac{\partial h(\bar{x})}{\partial x_j} \right) \left(\tilde{x}_k^{(i)} \frac{\partial h(\bar{x})}{\partial x_k} \right)^T \quad (14.48)$$

Now recall that $\tilde{x}_j^{(i)} = -\tilde{x}_j^{(i+n)}$ and $\tilde{x}_k^{(i)} = -\tilde{x}_k^{(i+n)}$ for $i = 1, \dots, n$. Therefore, the covariance approximation becomes

$$\begin{aligned}
P_u &= \frac{1}{n} \sum_{i=1}^n \sum_{j,k=1}^n \left(\tilde{x}_j^{(i)} \frac{\partial h(\bar{x})}{\partial x_j} \right) \left(\tilde{x}_k^{(i)} \frac{\partial h(\bar{x})}{\partial x_k} \right)^T \\
&= \sum_{j,k=1}^n P_{jk} \frac{\partial h(\bar{x})}{\partial x_j} \left(\frac{\partial h(\bar{x})}{\partial x_k} \right)^T \\
&= H P H^T \quad (14.49)
\end{aligned}$$

where the last equality comes from Equation (14.22). Comparing this equation for P_u with the true covariance of y from Equation (14.23), we see that Equation (14.43)

approximates the true covariance of y up to the third order (i.e., only terms to the fourth and higher powers are incorrect). This is the same approximation order as the linearization method, as seen on page 439. However, we would intuitively expect the magnitude of the error of the unscented approximation in Equation (14.43) to be smaller than the linear approximation HPH^T , because the unscented approximation at least contains correctly signed terms to the fourth power and higher, whereas the linear approximation does not contain any terms other than HPH^T .

The unscented transformation can be summarized as follows.

The unscented transformation

1. We begin with an n -element vector x with known mean $\bar{x}$ and covariance P . Given a known nonlinear transformation $y = h(x)$, we want to estimate the mean and covariance of y , denoted as $\bar{y}_u$ and P_u .
2. Form $2n$ sigma point vectors $x^{(i)}$ as follows:

$$\begin{aligned} x^{(i)} &= \bar{x} + \tilde{x}^{(i)} \quad i = 1, \dots, 2n \\ \tilde{x}^{(i)} &= \left(\sqrt{nP} \right)_i^T \quad i = 1, \dots, n \\ \tilde{x}^{(n+i)} &= - \left(\sqrt{nP} \right)_i^T \quad i = 1, \dots, n \end{aligned} \quad (14.50)$$

where $\sqrt{nP}$ is the matrix square root of nP such that $(\sqrt{nP})^T \sqrt{nP} = nP$, and $(\sqrt{nP})_i$ is the i th row of $\sqrt{nP}$.

3. Transform the sigma points as follows:

$$y^{(i)} = h(x^{(i)}) \quad i = 1, \dots, 2n \quad (14.51)$$

4. Approximate the mean and covariance of y as follows:

$$\begin{aligned} \bar{y}_u &= \frac{1}{2n} \sum_{i=1}^{2n} y^{(i)} \\ P_u &= \frac{1}{2n} \sum_{i=1}^{2n} \left(y^{(i)} - \bar{y}_u \right) \left(y^{(i)} - \bar{y}_u \right)^T \end{aligned} \quad (14.52)$$

■ EXAMPLE 14.1

To illustrate the unscented transformation, consider the nonlinear transformation shown in Equation (14.1). Since there are two independent variables (r and θ), we have $n = 2$. The covariance of P is given as $P = \text{diag}(\sigma_r^2, \sigma_\theta^2)$. Equation (14.32) shows that $W^{(i)} = 1/4$ for $i = 1, 2, 3, 4$. Equation (14.29) shows that the sigma points are determined as

$$\begin{aligned} x^{(1)} &= \bar{x} + \left(\sqrt{nP} \right)_1^T \\ &= \begin{bmatrix} 1 + \sigma_r \sqrt{2} \\ \pi/2 \end{bmatrix} \end{aligned}$$